

DARSHIL VAJA

Ahmedabad, Gujarat, India | +91 96624 90912 | darshil1293@gmail.com

[LinkedIn](#) | [GitHub](#)

Summary

Machine Learning and AI enthusiast with hands-on experience building end-to-end ML, Computer Vision, and Generative AI projects. Skilled in Python, Scikit-Learn, Pandas, NumPy, OpenCV, YOLOv8, Sentence Transformers, ChromaDB, FastAPI, Streamlit, model evaluation, and GitHub-based project documentation. Experienced in developing production-style projects involving data preprocessing, feature engineering, vector search, API development, testing, and deployment-ready structure. Seeking entry-level roles as a Machine Learning Engineer, AI/ML Engineer, or Data Science Intern.

Technical Skills

- **Programming:** Python, SQL, C, C++, Java, HTML, CSS
 - **Computer Science:** Data Structures and Algorithms, Operating Systems, Object Oriented Programming, Database Management System.
 - **Software Engineering:** Debugging, Basic Unit Testing, Code Reviews, Error Handling, Documentation, Agile Models.
 - **Machine Learning:** Supervised Learning, Regression, Classification, Feature Engineering, Model Evaluation, Threshold Tuning
 - **Libraries & Frameworks:** Pandas, NumPy, Scikit-Learn, Matplotlib, OpenCV, YOLOv8, EasyOCR
 - **Tools & Platforms:** Git, GitHub, Jupyter Notebook, VS Code, Kaggle
 - **Core Concepts:** Data Preprocessing, EDA, Train-Test Split, Imbalanced Classification, Computer Vision, OCR
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Projects

RAG-based Document Q&A System | [Github](#)

Tech stack: Python, Sentence Transformers, ChromaDB, FastAPI, Streamlit, pytest, Docker, GitHub Actions

- Built a production-style Retrieval-Augmented Generation system that allows users to upload PDF documents and ask source-grounded questions.
- Implemented PDF parsing, text cleaning, metadata-aware chunking, Sentence Transformer embeddings, ChromaDB vector storage, Cross-Encoder re-ranking, and fallback handling.
- Developed a Streamlit chat interface and FastAPI backend with upload, ask, health, and reset endpoints, along with pytest tests, Docker support, and GitHub Actions CI.

Credit Card Fraud Detection using Machine Learning | [GitHub](#)

Tech stack: Python, Scikit-Learn, Pandas, NumPy, Matplotlib

- Built an end-to-end computer vision system to detect vehicle number plates from images using a custom-trained YOLOv8 model.
- Integrated EasyOCR to extract text from cropped plate regions and saved detected images, crops, and CSV prediction outputs.

- Implemented single-image and batch prediction workflows with a clean GitHub-ready folder structure and documentation.

Vehicle Number Plate Recognition using YOLOv8 | [GitHub](#)

Tech stack: Python, YOLOv8, OpenCV, EasyOCR, Pandas

- Developed an ML classification pipeline to identify genuine and fraudulent transactions from an imbalanced dataset.
- Compared Logistic Regression, class-weighted Logistic Regression, and Random Forest using precision, recall, F1-score, ROC-AUC, PR-AUC, and confusion matrix analysis.
- Applied threshold tuning and built a raw transaction prediction workflow for testing model predictions on new samples.

Customer Churn Prediction using Machine Learning | [GitHub](#)

Tech stack: Python, Scikit-Learn, FastAPI, pytest, GitHub Actions

- Built a classification pipeline to predict customer churn using structured customer data.
- Performed data preprocessing, feature encoding, model training, and evaluation using classification metrics such as accuracy, precision, recall, F1-score, and confusion matrix.
- Added FastAPI integration, pytest-based tests, and GitHub Actions workflow to make the project more production-oriented and portfolio-ready.

Experience

Python Web Developer Intern TECHInfinity	Feb 2026 – Apr 2026
• Worked on Python-based development tasks and contributed to data handling, dashboard workflow, and backend logic. • Improved practical understanding of structured coding, debugging, project documentation, and professional task execution.	

Education

B.Tech. Computer Science and Technology	2022 - 2026
Specialization: Artificial Intelligence and Machine Learning	
Unitedworld Institute of Technology, Karnavati University	
Higher Secondary School	2021
Dalmia Public School, CBSE CGPA: 8.1	
Secondary School	2019
Dalmia Public School, CBSE CGPA: 7.8	
